

ICT600 CORRECTED STUDENT SCHEME

January 2024

Question-ordered model answers and marking guide

Scope	January 2024 final examination
Arrangement	Answers follow the original paper from Question 1 through Part B. Coding, diagrams, and written responses are integrated once at their correct positions.
Assessment	100 marks: Part A 70, Part B 30
Status	Corrected student reference; equivalent technically correct answers may receive credit.

How to use this scheme

Question-order rule

Attempt the paper first. Then check the matching answer in this document. Each answer appears once, in the same order as the question paper.

- Equivalent correct wording, diagrams, algorithms, and code may receive full credit when they meet the stated requirements.
- For coding, award method marks when a minor syntax error does not invalidate an otherwise correct solution.
- For design or illustration questions, the labelled diagram is the model answer; prose alone is not treated as a complete design.

Part A

Question 1(a)

8 marks

Describe Intranet, Extranet, Internet, and LAN.

Mark allocation

Awardable element	Marks
Intranet	2
Extranet	2
Internet	2
LAN	2
Total	8

Complete model answer

- Intranet: a private organisational network for authorised internal users.
- Extranet: controlled access to selected organisational resources for authorised external parties.
- Internet: the global network of interconnected networks using standard protocols.
- LAN: a network covering a limited area such as a room, building, or campus section.

Question 1(b)

10 marks

Identify four cloud applications for UiTM teaching and learning, with examples.

Mark allocation

Awardable element	Marks
First application and use	2.5
Second application and use	2.5
Third application and use	2.5
Fourth application and use	2.5
Total	10

Complete model answer

- Any four suitable applications, such as UFuture, Google Classroom, Microsoft Teams, Google Meet, Google Drive, or OneDrive, each paired with an accurate teaching use.

Question 2(a) - HTML table

6 marks

Write complete HTML for Table 1: List of Names.

Mark allocation

Awardable element	Marks
Table/caption structure	1
Three headings	1
Three complete data rows	3
Valid nesting and closing tags	1
Total	6

Complete model answer

```
<!doctype html>
<html lang="en">
<head><meta charset="utf-8"><title>List of Names</title></head>
<body>
<table border="1">
  <caption>Table 1: List of Names</caption>
  <tr><th>Firstname</th><th>Lastname</th><th>Age</th></tr>
  <tr><td>Jill</td><td>Smith</td><td>50</td></tr>
  <tr><td>Eve</td><td>Jackson</td><td>94</td></tr>
  <tr><td>John</td><td>Doe</td><td>80</td></tr>
</table>
</body>
</html>
```

Question 2(b) - styled HTML page

6 marks

Write the complete page with the required background, alignment and colours.

Mark allocation

Awardable element	Marks
Valid HTML and required text	2
Powderblue body and centred text	2
Blue h1	1
Red paragraph	1
Total	6

Complete model answer

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <title>Web Application and Technology</title>
  <style>
    body { background-color: powderblue; text-align: center; }
    h1 { color: blue; }
    p { color: red; }
  </style>
</head>
<body>
  <h1>Web Application and Technology</h1>
  <p>Explore web technology and application</p>
</body>
</html>
```

Question 3(a) - JavaScript variables and output

8 marks

Store the four printed values in separate variables and reproduce the output.

Mark allocation

Awardable element	Marks
Four correct variables/values	4
Four correct browser outputs	4
Total	8

Complete model answer

```
<!doctype html>
<html lang="en"><body>
<p id="name"></p><p id="program"></p><p id="part"></p><p id="cgpa"></p>
<script>
const studentName = "Imran bin Haziq";
const programCode = "CS240";
const part = 5;
const cgpa = 3.55;

document.getElementById("name").textContent = "Name: " + studentName;
document.getElementById("program").textContent = "Program Code: " + programCode;
document.getElementById("part").textContent = "Part: " + part;
document.getElementById("cgpa").textContent = "CGPA: " + cgpa;
</script>
</body></html>
```

Question 3(b) - displayUserName

10 marks

Write and call displayUserName('Aidan') to display Username: Aidan.

Mark allocation

Awardable element	Marks
Function declaration	2
Parameter	2
Username prefix and concatenation	2
Browser output	2
Correct call	2
Total	10

Complete model answer

```
<!doctype html>
<html lang="en"><body>
<p id="result"></p>
<script>
function displayUserName(userName) {
    document.getElementById("result").textContent = "Username: " + userName;
}
displayUserName("Aidan");
</script>
</body></html>
```

Expected output / behaviour

Username: Aidan

Question 4(a) - PHP multidimensional array

4 marks

Use a PHP multidimensional array to display LN11/London and OX47/Oxford.

Mark allocation

Awardable element	Marks
Correct nested array	2
Both rows displayed in a table	2
Total	4

Complete model answer

```
<?php
$scities = [
    ['LN11', 'London'],
    ['OX47', 'Oxford'],
];
?>
<table border="1">
<?php foreach ($scities as $city): ?>
    <tr><td><?=$city[0] ?></td><td><?=$city[1] ?></td></tr>
<?php endforeach; ?>
</table>
```

Question 4(b)

6 marks

Trace the supplied PHP string script.

Mark allocation

Awardable element	Marks
Length and original output	1
Uppercase	1
Lowercase	1
Replacement	1
Substring	1
Both strpos values and types	1
Total	6

Complete model answer

- 43
- How can I cook rice in a clean round pot?
- HOW CAN I COOK RICE IN A CLEAN ROUND POT?
- how can i cook rice in a clean round pot?
- How round I cook rice in a clean round pot?
- w roun
- bool(false)
- bool(false)

Marker note

The final result is bool(false), not int(4), because 'could' is absent after the replacement.

Question 5(a) - flood table

6 marks

Create Table 3 exactly with the spanning title and all values.

Mark allocation

Awardable element	Marks
Table structure	1
colspan title row	2
Location headings	1
Complete data rows	2
Total	6

Complete model answer

```
<table border="1">
  <caption>Table 3: Mangsa Banjir di Pahang</caption>
  <tr><th colspan="2">Bilangan Mangsa Banjir Pahang 2021</th></tr>
  <tr><th>Temerloh</th><th>Kuantan</th></tr>
  <tr><td>900 orang</td><td>3314 orang</td></tr>
  <tr><td>253 keluarga</td><td>892 keluarga</td></tr>
</table>
```

Question 5(b) - HTML and JavaScript interaction

6 marks

Build the illustrated page and make the button update its message.

Mark allocation

Awardable element	Marks
Heading, paragraph and button	3
Event handler	1
Correct DOM lookup and content update	2
Total	6

Complete model answer

```
<!doctype html>
<html lang="en"><body>
  <h1>What Can JavaScript Do?</h1>
  <p id="message">Hello JavaScript!</p>
  <button type="button" id="changeMessage">Click Me!</button>
  <script>
    document.getElementById("changeMessage").addEventListener("click", () => {
      document.getElementById("message").textContent = "JavaScript can change HTML content!";
    });
  </script>
</body></html>
```

Part B

Part B Question 1

15 marks

Construct a checkout-payment wireframe for an e-commerce website.

Mark allocation

Awardable element	Marks
Header and store identity	2
Navigation and checkout page path	2
Billing details	2
Delivery details and method	2
Payment method	2
Order summary and total	2
Confirmation action	2
Footer and coherent layout	1
Total	15

Complete model answer

- The model below includes the required billing, delivery, payment, header, footer, page-name, navigation, order-summary, and confirmation areas.

HEADER: LOGO | SEARCH | ACCOUNT / CART

NAVIGATION > CART > CHECKOUT > PAYMENT

BILLING ADDRESS

DELIVERY ADDRESS

DELIVERY METHOD

ORDER SUMMARY
Items | Delivery | Total

PAYMENT METHOD
Card / Online Banking

PLACE ORDER

FOOTER: CONTACT | DELIVERY | RETURNS | PRIVACY

Part B Question 2 - PHP user registration

15 marks

Provide the complete form, POST validation, success state and error state.

Mark allocation

Awardable element	Marks
Form and required/optional controls	4
POST request handling	2
Required-field validation	3
Success and failure messages	2
Safe encoded output	2
Valid overall structure	2
Total	15

Complete model answer

```
<?php
$values = ['fullname' => '', 'email' => '', 'birthdate' => '', 'gender' => ''];
$message = '';

function e(string $value): string {
    return htmlspecialchars($value, ENT_QUOTES, 'UTF-8');
}

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    foreach ($values as $key => $unused) {
        $values[$key] = trim($_POST[$key] ?? '');
    }
    $password = $_POST['password'] ?? '';
    if ($values['fullname'] === '' || $values['email'] === '' || $password === '') {
        $message = 'Unsuccessful: enter all required information.';
    } else {
        $message = 'Successful signup for ' . $values['fullname'] . '.';
    }
}
?>
<!doctype html>
<html lang="en"><head><meta charset="utf-8"><title>User Signup</title></head>
<body>
<h1>User Signup</h1>
<form method="post">
    <label>Full name* <input name="fullname" value="<?= e($values['fullname']) ?>" required></label><br>
    <label>Email* <input type="email" name="email" value="<?= e($values['email']) ?>" required></label><br>
    <label>Password* <input type="password" name="password" required></label><br>
    <label>Birthdate <input type="date" name="birthdate" value="<?= e($values['birthdate']) ?>"></label><br>
    <label><input type="radio" name="gender" value="M"> Male</label>
    <label><input type="radio" name="gender" value="F"> Female</label><br>
    <button type="submit">Signup</button>
</form>
<?php if ($message !== ''): ?><p><?= e($message) ?></p><?php endif; ?>
</body></html>
```

Marker note

The stray 22-mark label in the historical official answer is incorrect; the printed question is 15 marks.